

Technical note: exactness, inference, and gates

Versioned implementation note for independent review

1. Computational authority

The packed hot state is authoritative for migrated physical, organizational, logistics, and action calculations. Static topology is immutable. Sparse reference processes cross an explicit synchronization boundary. Structural rebuilds materialize all lanes only when dynamic entity IDs change. This division prevents stale packed values from overwriting sparse-event mutations.

2. Exactness result

The exactness battery compares component hashes for the decision state and a continuation payload that includes scheduler/RNG lineage. It uses 12 preregistered seeds and five horizons: 0.25, 0.5, 1, 2, and 7 days. The resampling test forces a known parent distribution, checks systematic parent indices and independent child lineages, then verifies continuation divergence.

Component	Observed
Exactness comparisons	True
Packed authority inventory	True
Kernel oracle battery	True
Profile accounting	packed_state_and_synchronization

3. Synthetic inference validation

The passing v2 protocol is synthetic and history-free. It includes Rao-Blackwellized filtering, Gaussian guided proposals, MCSE reporting, and a combined gate. It reports coverage, RMSE scaled by posterior standard deviation, variance recovery, ESS ratio, and p95 error. The earlier v1 guided validation failed its ESS-ratio threshold and remains preserved as a negative result; v2 changed the synthetic observation/proposal to make the validation informative without changing thresholds.

Metric	Threshold family	Current status
RB coverage	predeclared coverage interval	PASS
Guided coverage / RMSE	coverage and normalized RMSE	PASS in v2; v1 preserved as failed
MCSE	finite and within predeclared bound	PASS
Combined	all components pass	PASS in v2

4. Performance and migration

The representative cProfile run is intentionally smaller than the fixed acceptance workload: four particles for seven days, three repeats. Its dominant self-time domain is packed-state synchronization and serialization, followed by packed execution. The next migration target is therefore row materialization and lane synchronization, not blind conversion of sparse physical or organizational processes.

Profile quantity	Median
Profile CPU seconds	38.43281210000001
Packed synchronization self seconds	16.242639700000005
Packed execution self seconds	6.3854225
Sparse reference self seconds	0.09408439999999998

The separate fixed 32-particle x 8-week x 3-branch production run did not complete within the controlled resource window. It produced no partial timing result. This is recorded as an implementation/deployment failure, not as evidence for or against the scientific mechanism.

5. Historical authorization rule

Historical work requires a passing exactness/performance/inference spine and a clean core boundary. The contract records the current model hash and all dirty paths. Existing historical v5 outputs are never silently rebound to a new model hash. A current-core historical confrontation is once-only: no calibration loop, no post-hoc branch selection, and no deletion of negative seeds.

6. Independent reproduction command family

Run the phase scripts in order: exactness battery, authority audit, kernel oracles, representative profile, fixed-work benchmark, phase-A certificate, phase-C contract, phase-D historical gate, phase-E theory contract, phase-F robustness status, then packet builder. Every script refuses to overwrite an existing evidence artifact.